# MSG No. 135.494 $P_I4_2/mbc$ [ Type IV, tetragonal ]

Table 1: Wyckoff site: `4a`, site symmetry: `4'2'2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, \frac{1}{4}\right]$ | `[1,6,7,8,18,19,20,21]` |
| 2 | $\left[0, \frac{1}{2}, \frac{3}{4}\right]$ | `[2,3,4,5,17,22,23,24]` |
| 3 | $\left[\frac{1}{2}, 0, \frac{3}{4}\right]$ | `[9,14,15,16,26,27,28,29]` |
| 4 | $\left[0, \frac{1}{2}, \frac{1}{4}\right]$ | `[10,11,12,13,25,30,31,32]` |

Table 2: Wyckoff site: `4b`, site symmetry: `-42'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$ | `[1,6,10,11,20,21,31,32]` |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$ | `[2,3,9,14,23,24,28,29]` |
| 3 | $\left[0, 0, \frac{3}{4}\right]$ | `[4,5,15,16,17,22,26,27]` |
| 4 | $\left[0, 0, \frac{1}{4}\right]$ | `[7,8,12,13,18,19,25,30]` |

Table 3: Wyckoff site: `4c`, site symmetry: `4'/m..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, 0\right]$ | `[1,6,9,14,18,19,26,27]` |
| 2 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | `[2,3,10,11,17,22,25,30]` |
| 3 | $\left[0, \frac{1}{2}, 0\right]$ | `[4,5,12,13,23,24,31,32]` |
| 4 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[7,8,15,16,20,21,28,29]` |

Table 4: Wyckoff site: `4d`, site symmetry: `m.m'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[1,6,9,14,23,24,31,32]` |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[2,3,10,11,20,21,28,29]` |
| 3 | $[0, 0, 0]$ | `[4,5,12,13,18,19,26,27]` |
| 4 | $\left[0, 0, \frac{1}{2}\right]$ | `[7,8,15,16,17,22,25,30]` |

Table 5: Wyckoff site: `8e`, site symmetry: `..2/m'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[1,7,25,31]` |
| 2 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[2,5,26,29]` |
| 3 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[3,4,27,28]` |
| 4 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[6,8,30,32]` |
| 5 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[9,15,17,23]` |
| 6 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[10,13,18,21]` |
| 7 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[11,12,19,20]` |

| | | |
|---|---|---|
| 8 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | [14,16,22,24] |

Table 6: Wyckoff site: 8f, site symmetry: 4'..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, z\right]$ | [1,6,18,19] |
| 2 | $\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$ | [2,3,17,22] |
| 3 | $\left[0, \frac{1}{2}, -z\right]$ | [4,5,23,24] |
| 4 | $\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$ | [7,8,20,21] |
| 5 | $\left[\frac{1}{2}, 0, -z\right]$ | [9,14,26,27] |
| 6 | $\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$ | [10,11,25,30] |
| 7 | $\left[0, \frac{1}{2}, z\right]$ | [12,13,31,32] |
| 8 | $\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$ | [15,16,28,29] |

Table 7: Wyckoff site: 8g, site symmetry: 2.m'm'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, \frac{1}{2}, z\right]$ | [1,6,31,32] |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$ | [2,3,28,29] |
| 3 | $[0, 0, -z]$ | [4,5,26,27] |
| 4 | $\left[0, 0, \frac{1}{2}-z\right]$ | [7,8,25,30] |
| 5 | $\left[\frac{1}{2}, \frac{1}{2}, -z\right]$ | [9,14,23,24] |
| 6 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-z\right]$ | [10,11,20,21] |
| 7 | $[0, 0, z]$ | [12,13,18,19] |
| 8 | $\left[0, 0, z+\frac{1}{2}\right]$ | [15,16,17,22] |

Table 8: Wyckoff site: 8h, site symmetry: m.2'm'

| No. | position | mapping |
|---|---|---|
| 1 | $[x, x, 0]$ | [1,14,23,32] |
| 2 | $\left[-x, x, \frac{1}{2}\right]$ | [2,11,21,28] |
| 3 | $\left[x, -x, \frac{1}{2}\right]$ | [3,10,20,29] |
| 4 | $\left[x+\frac{1}{2}, \frac{1}{2}-x, 0\right]$ | [4,13,19,26] |
| 5 | $\left[\frac{1}{2}-x, x+\frac{1}{2}, 0\right]$ | [5,12,18,27] |
| 6 | $[-x, -x, 0]$ | [6,9,24,31] |
| 7 | $\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$ | [7,16,17,30] |
| 8 | $\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}\right]$ | [8,15,22,25] |

Table 9: Wyckoff site: 16i, site symmetry: ..2

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$ | [1,7] |
| 2 | $\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$ | [2,5] |
| 3 | $\left[x+\frac{1}{2}, -x, \frac{3}{4}\right]$ | [3,4] |

*continued ...*

Table 9

| No. | position | mapping |
|---|---|---|
| 4 | $\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$ | [6,8] |
| 5 | $\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$ | [9,15] |
| 6 | $\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$ | [10,13] |
| 7 | $\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$ | [11,12] |
| 8 | $\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$ | [14,16] |
| 9 | $\left[x+\frac{1}{2},\ x,\ \frac{3}{4}\right]$ | [17,23] |
| 10 | $\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$ | [18,21] |
| 11 | $\left[x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$ | [19,20] |
| 12 | $\left[\frac{1}{2}-x,\ -x,\ \frac{3}{4}\right]$ | [22,24] |
| 13 | $\left[\frac{1}{2}-x,\ -x,\ \frac{1}{4}\right]$ | [25,31] |
| 14 | $\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$ | [26,29] |
| 15 | $\left[-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$ | [27,28] |
| 16 | $\left[x+\frac{1}{2},\ x,\ \frac{1}{4}\right]$ | [30,32] |

Table 10: Wyckoff site: 16j, site symmetry: .2'.

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 0,\ \frac{1}{4}\right]$ | [1,20] |
| 2 | $\left[0,\ x,\ \frac{3}{4}\right]$ | [2,23] |
| 3 | $\left[0,\ -x,\ \frac{3}{4}\right]$ | [3,24] |
| 4 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$ | [4,17] |
| 5 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$ | [5,22] |
| 6 | $\left[-x,\ 0,\ \frac{1}{4}\right]$ | [6,21] |
| 7 | $\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$ | [7,18] |
| 8 | $\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$ | [8,19] |
| 9 | $\left[-x,\ 0,\ \frac{3}{4}\right]$ | [9,28] |
| 10 | $\left[0,\ -x,\ \frac{1}{4}\right]$ | [10,31] |
| 11 | $\left[0,\ x,\ \frac{1}{4}\right]$ | [11,32] |
| 12 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$ | [12,25] |
| 13 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$ | [13,30] |
| 14 | $\left[x,\ 0,\ \frac{3}{4}\right]$ | [14,29] |
| 15 | $\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$ | [15,26] |
| 16 | $\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$ | [16,27] |

Table 11: Wyckoff site: 16k, site symmetry: m..

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ 0]$ | [1,14] |
| 2 | $\left[-y,\ x,\ \frac{1}{2}\right]$ | [2,11] |
| 3 | $\left[y,\ -x,\ \frac{1}{2}\right]$ | [3,10] |
| 4 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$ | [4,13] |
| 5 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$ | [5,12] |
| 6 | $[-x,\ -y,\ 0]$ | [6,9] |
| 7 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | [7,16] |

*continued ...*

Table 11

| No. | position | mapping |
|---|---|---|
| 8 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | [8,15] |
| 9 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | [17,30] |
| 10 | $\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ 0\right]$ | [18,27] |
| 11 | $\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$ | [19,26] |
| 12 | $\left[x,\ -y,\ \frac{1}{2}\right]$ | [20,29] |
| 13 | $\left[-x,\ y,\ \frac{1}{2}\right]$ | [21,28] |
| 14 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | [22,25] |
| 15 | $[y,\ x,\ 0]$ | [23,32] |
| 16 | $[-y,\ -x,\ 0]$ | [24,31] |

Table 12: Wyckoff site: 16l, site symmetry: ..m'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ x,\ z]$ | [1,32] |
| 2 | $\left[-x,\ x,\ z+\frac{1}{2}\right]$ | [2,28] |
| 3 | $\left[x,\ -x,\ z+\frac{1}{2}\right]$ | [3,29] |
| 4 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$ | [4,26] |
| 5 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ -z\right]$ | [5,27] |
| 6 | $[-x,\ -x,\ z]$ | [6,31] |
| 7 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [7,30] |
| 8 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | [8,25] |
| 9 | $[-x,\ -x,\ -z]$ | [9,24] |
| 10 | $\left[x,\ -x,\ \frac{1}{2}-z\right]$ | [10,20] |
| 11 | $\left[-x,\ x,\ \frac{1}{2}-z\right]$ | [11,21] |
| 12 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z\right]$ | [12,18] |
| 13 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$ | [13,19] |
| 14 | $[x,\ x,\ -z]$ | [14,23] |
| 15 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | [15,22] |
| 16 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [16,17] |

Table 13: Wyckoff site: 32m, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | [1] |
| 2 | $\left[-y,\ x,\ z+\frac{1}{2}\right]$ | [2] |
| 3 | $\left[y,\ -x,\ z+\frac{1}{2}\right]$ | [3] |
| 4 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$ | [4] |
| 5 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$ | [5] |
| 6 | $[-x,\ -y,\ z]$ | [6] |
| 7 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [7] |
| 8 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | [8] |
| 9 | $[-x,\ -y,\ -z]$ | [9] |
| 10 | $\left[y,\ -x,\ \frac{1}{2}-z\right]$ | [10] |
| 11 | $\left[-y,\ x,\ \frac{1}{2}-z\right]$ | [11] |

*continued ...*

Table 13

| No. | position | mapping |
|---|---|---|
| 12 | $\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$ | [12] |
| 13 | $\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$ | [13] |
| 14 | $[x,\, y,\, -z]$ | [14] |
| 15 | $\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$ | [15] |
| 16 | $\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [16] |
| 17 | $\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [17] |
| 18 | $\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z\right]$ | [18] |
| 19 | $\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z\right]$ | [19] |
| 20 | $\left[x,\, -y,\, \frac{1}{2}-z\right]$ | [20] |
| 21 | $\left[-x,\, y,\, \frac{1}{2}-z\right]$ | [21] |
| 22 | $\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$ | [22] |
| 23 | $[y,\, x,\, -z]$ | [23] |
| 24 | $[-y,\, -x,\, -z]$ | [24] |
| 25 | $\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$ | [25] |
| 26 | $\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, -z\right]$ | [26] |
| 27 | $\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, -z\right]$ | [27] |
| 28 | $\left[-x,\, y,\, z+\frac{1}{2}\right]$ | [28] |
| 29 | $\left[x,\, -y,\, z+\frac{1}{2}\right]$ | [29] |
| 30 | $\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [30] |
| 31 | $[-y,\, -x,\, z]$ | [31] |
| 32 | $[y,\, x,\, z]$ | [32] |